

MCMB-HDT: A Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation

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Abstract—The emergence of a multi-band cellular landscape—spanning sub-6 GHz, millimeter-wave, and the upper mid-band (FR3)—presents significant challenges for handset design and network optimization. Handsets must support multiple antennas in a compact form factor. As the field of view of each antenna is limited, the antenna layout, search, and switching policies become paramount. This paper presents the Multi-Cell Multi-Band Handset Digital Twin (MCMB-HDT), a high-fidelity, UE-centric framework for generating time-resolved wireless data for studying these problem. By integrating geographically grounded urban scenes with Sionna-based ray tracing and detailed, multi-band handset antenna modeling, the MCMB-HDT captures joint multi-frequency statistics and rapid channel dynamics driven by pedestrian pose and mobility. We demonstrate the utility of this digital twin through a closed-loop array-activation use case, where a measurement-limited agent must select optimal antenna modules under strict power and observation constraints. Using Deep Reinforcement Learning (Proximal Policy Optimization), we show that policies trained and evaluated within the MCMB-HDT environment significantly outperform traditional greedy and bandit baselines. Our results quantify the performance impact of handset embodiment and local dynamics, providing a robust substrate for the design of future application-aware, closed-loop wireless protocols.

Index Terms—Digital twin networks, user equipment modeling, handset antenna arrays, multi-cell multi-band wireless systems, ray tracing, closed-loop control, reinforcement learning.

I. INTRODUCTION

The next-generation cellular landscape is emerging as a complex patchwork of frequencies, spanning traditional sub-6 GHz bands, millimeter-wave (mmWave), and the increasingly critical upper mid-band (FR3) [1]–[3]. While this multi-band operation promises improved coverage from macro-diversity, greater spectrum, and enhanced resilience, supporting multiple frequencies imposes severe constraints on the mobile handset. Current 3GPP specifications already define over 100 band combinations [4], and this number is expected to grow as FR3 integration matures [5], [6].

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Supporting this diversity of frequencies on a portable device is a multi-dimensional challenge: real estate for antennas is scarce, power budgets are rigid, and the physical interaction between the handset and the user’s body introduces significant signal volatility. Optimizing these systems requires more than isolated link-level analysis; it demands *Digital Twins* (DTs) capable of capturing the high-fidelity interplay between hardware embodiment, site-specific propagation, and real-time dynamics across the protocol stack. However, current DT methodologies face four critical bottlenecks:

Handset Embodiment & User Interaction: To begin, multi-band handsets are themselves challenging to model as radiation patterns depend in complex ways on the antenna types, layout, and materials of the handset [5], [7]–[9]. The radiation patterns are also strongly influenced by how the handset is held as the hand and body can significantly block signals, particularly in higher frequencies in the upper mid-band and mmWave range [10], [11].

Joint Multi-Frequency Statistics: Multi-frequency simulations require capturing joint statistics of the large-scale propagation across multiple frequencies to assess functions as handover, macro-diversity, beam tracking, and channel tracking [1]–[3]. As reflectivity and transmission are strongly frequency-dependent [12], locations of buildings, blockers, and their material properties influence different frequencies in different manners.

Spatiotemporal Motion Dynamics: Motion in these environments can lead to rapid changes in channel characteristics – particularly in high frequencies where diffraction is limited, and transitions around obstacles become much faster [13], [14]. In addition, due to the limited field-of-view of each antenna module, changes in orientation of the handset can also lead to high variability in channel quality. These dynamics need to be captured for the design and evaluation of many protocols including channel tracking, rate prediction, beam tracking, and even higher layer protocols such as congestion control [15], [16].

The Closed-Loop Requirement: Control decisions on mobile devices such as antenna or beam selection, influence channel and/or their measurements. Thus, UE-side control algorithms cannot thus be designed in an “open-loop” fashion; the DT must serve as a reactive substrate for training and testing control policies.

Our Approach: To address these challenges, this paper presents a *multi-cell multi-band handset digital twin*, abbreviated as MCMB-HDT (shown in Fig. 1). The DT does not

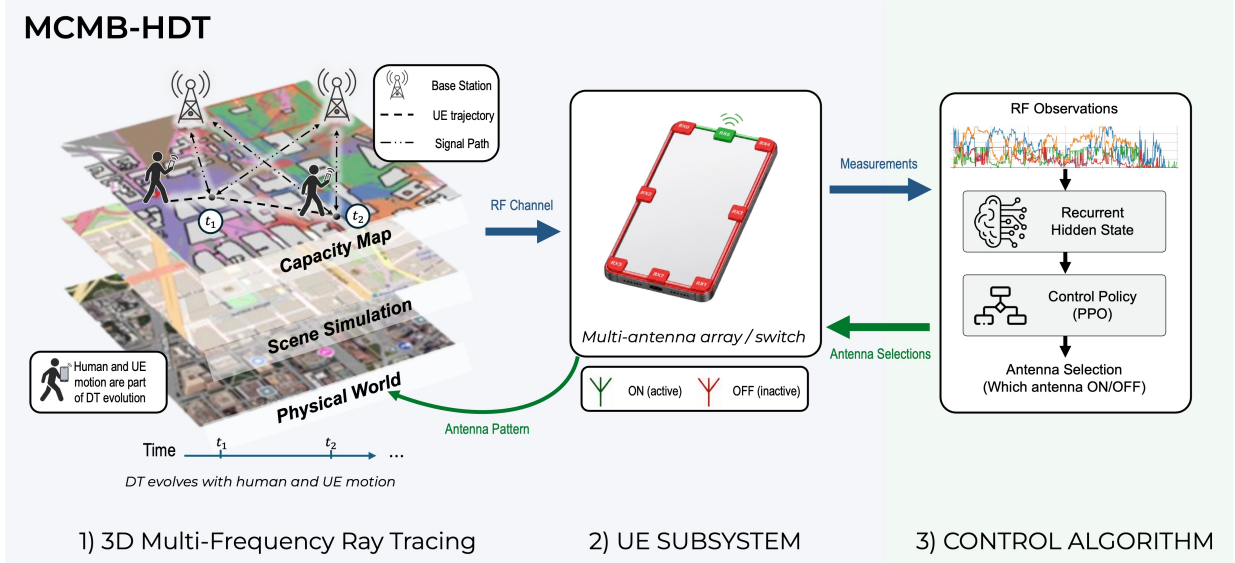


Fig. 1. Overview of the proposed MCMB-HDT and its closed-loop array-activation use case. The left panel illustrates the UE-centric digital-twin pipeline, where a geographically grounded 3D urban scene, pedestrian UE trajectory, and handset pose dynamics are combined with multi-frequency ray tracing to generate time-varying channel, CQI, and rate traces. The middle panel shows the embodied multi-array handset, whose active and inactive antenna states determine the effective receive pattern exposed to the RF channel. The right panel shows the measurement-limited control loop: RF observations from the active arrays update a recurrent hidden state, a PPO-based policy selects the next antenna ON/OFF configuration, and the resulting antenna stimulus feeds back to the handset and digital twin for subsequent measurements.

twin the base-station topology or UE device in isolation. Instead, it twins the interaction among a pedestrian-carried handset, the surrounding urban geometry, and the available wireless infrastructure. Concretely, the MCMB-HDT combines a geographically grounded urban scene generated from real map data [17], Sionna-based ray tracing [18], a detailed configurable, handset platform spanning FR1 and FR3, and stochastic route and pose dynamics for a pedestrian-carried device. The output of this digital twin is a set of time-varying CQI and rate traces across feasible cell-array links. The framework also supports controlled ablations that isolate the impact of building fidelity, handset embodiment, and local pose dynamics on the generated wireless data.

To demonstrate how this digital twin can be used in practice, we instantiate a measurement-limited closed-loop array-activation problem on top of the DT-generated traces. For this problem, we design two primary reinforcement learning (RL) policies based on Proximal Policy Optimization (PPO) [19], namely rate-aware and risk-aware variants, and compare them against simpler greedy and bandit baselines. This application is not the only role of the MCMB-HDT. Rather, it serves as a representative example showing that the digital twin is not merely a visualization or propagation-analysis tool. It provides a closed-loop experimental substrate on which wireless control policies can be designed, compared, and interpreted.

A. Our Contributions

The main contributions of this work are as follows.

UE-centric multi-cell multi-band handset digital twin. We develop the MCMB-HDT, a high-fidelity UE-centric digital twin that integrates real urban geometry, gNB configuration, detailed handset embodiment, on-device array placement,

band-dependent radiation patterns, and stochastic pedestrian and handset-pose dynamics into a unified wireless data-generation pipeline. Within the same DT framework, we demonstrate time-resolved data generation under different UE mobility levels and conduct controlled ablations on the building model, handset model, and local pose dynamics to isolate the effect of each fidelity component. This construction and DT-level analysis are presented in Section II.

Closed-loop array activation over DT-generated traces. We formulate a measurement-limited closed-loop array-activation problem on top of the CQI and rate traces generated by the MCMB-HDT. We then develop two primary PPO-based RL policies, denoted PPO-rate and PPO-risk, and compare them against greedy and bandit baselines as a concrete closed-loop use case enabled by the digital twin. The formulation and policy design are presented in Section III.

DT-enabled closed-loop performance evaluation. We use the MCMB-HDT-generated traces as an experimental substrate for evaluating measurement-limited closed-loop control policies. The experiments combine representative trajectory-level analysis and aggregate policy comparison to show how learning-based array activation behaves under partial RF observation. These results are presented in Section IV.

Related Work

DTs with ray tracing have becoming increasingly widely-used and remain the only viable data source for complex ML training. Channel and ray-tracing studies provide valuable propagation realism, large-scale sensing and communication datasets provide important empirical coverage [20], and wireless digital-twin frameworks have begun to support network-side simulation and task-specific decision making [21]–[23].

Most existing wireless DT workflows remain oriented toward network monitoring, planning, or static evaluation. They do not provide a high-fidelity UE-centric data-generation framework for adaptive closed-loop control. In particular, they do not jointly capture real urban geometry, serving multi-cell infrastructure, multi-band on-device arrays, handset-body effects, and stochastic pose dynamics in a form that can directly drive sequential wireless decisions.

Measurement-driven studies with form-factor UEs at 28 GHz show that hand/body occlusion and device rotation dominate coverage and beam management, challenging simplified blockage models [24], [25]. Antenna-layout studies for upper mid-bands further show that discrete and modular UE arrays create practical trade-offs in aperture placement and link robustness [9]. Multi-band campaigns (11/16/28/32 GHz) quantify ~ 20 dB human-blockage loss and validate KED/UTD abstractions [26], [27], while FR1/FR3 results at 6.75/16.95 GHz provide calibrated outdoor models and indoor-hotspot angular-spread statistics [28], [29].

Modeling tools, calibration pipelines, and wireless DTs provide the simulation substrate needed to reproduce these effects beyond isolated measurements. EM/RT pipelines benchmark KED/UTD/PO against 60 GHz measurements to expose accuracy–complexity trade-offs [30], while device-as-scatterer CAD and realistic 3GPP UE antenna models show that handset geometry and array placement cannot be reduced to fixed-angle masks [8], [31]. Site-specific calibration/validation of NYURay at upper mid-band tightens agreement with measurements, and NYUSIM offers spatially consistent, blockage-aware corpora for synthesis and analysis [32], [33]. Recent DT and dataset efforts further connect ray tracing, network simulation, sensing, and learning, including DeepSense 6G, Sionna-RT/ns-3 digital network twins, and DT-enhanced wireless navigation [20], [21], [23].

These modeling and DT substrates have enabled learning-based wireless prediction and control. AE+LSTM sequence models improve Top- C ranking and reduce misalignment for next-slot beam/BS prediction [34]. Cross-band learning maps sub-6 GHz features to mmWave beam/blockage, and vision/semantics aid LoS/NLoS forecasting and candidate selection [35]–[37]. Physics-informed RL and symbolic priors have been used for zero-shot wireless navigation, while transformer-based rate prediction and likelihood-based posterior inference show how learned models can exploit structured wireless observations for proactive decision making [22], [38]–[40].

II. MULTI-CELL MULTI-BAND HANDSET DIGITAL TWIN

This section presents the MCMB-HDT used in this work. As illustrated in Fig. 1, the MCMB-HDT integrates the urban environment and gNB-cell configuration, the handset body platform and on-device array configuration, the route and pose dynamics of a pedestrian-carried handset, and the Sionna-based [18] ray-tracing engine into a unified pipeline for generating time-varying multi-band CQI and rate traces. A key observation is that, in this MCMB setting, the generated dataset depends not only on where the UE moves, but also on how the city and the handset are modeled: the urban geometry

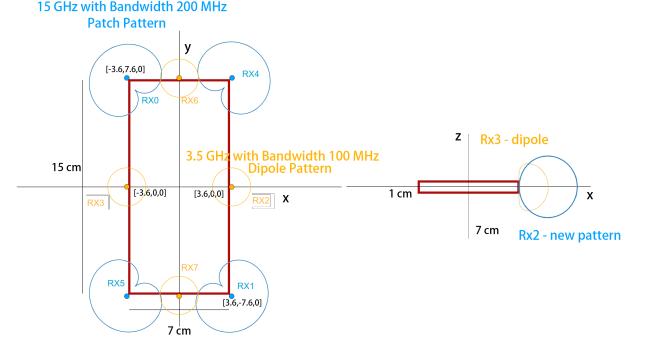


Fig. 2. Handset platform and on-device array configuration used in the MCMB-HDT. The red outline denotes the handset body of size $7\text{ cm} \times 15\text{ cm} \times 1\text{ cm}$. Four FR3 patch-type arrays with 15 GHz carrier frequency and 200 MHz bandwidth are placed near the four corners and are shown in blue. Four FR1 dipole-type arrays with 3.5 GHz carrier frequency and 100 MHz bandwidth are placed at the centers of the four edges and are shown in yellow. The left and right panels show the x - y and x - z views, respectively. For visual clarity in this DT section, only the representative subset RX0–RX3 is enabled in the illustrative figures based on this handset layout. The later closed-loop optimization uses the full eight-array handset configuration.

shapes blockage, reflection, and cell visibility, while the handset body, array placement, radiation patterns, and local self-rotation shape which links are exposed or attenuated over time. To make this dependence explicit, we later compare the full MCMB-HDT against several ablations, including removing the building model, removing the detailed handset body model, and fixing the handset local pose.

A. Handset Platform and Array Configuration

1) *Handset Platform Model*: The UE is not modeled as a point receiver. Instead, the handset is represented by a metallic rectangular body with dimensions $7\text{ cm} \times 15\text{ cm} \times 1\text{ cm}$, which serves both as the physical support for the receive arrays and as a potential self-blockage object in the ray-tracing simulation. This body-level modeling is important in the MCMB-HDT because the incident paths can be attenuated differently depending on which side of the handset is exposed to the serving cells and reflected components. The handset model is therefore equipped with its own local coordinate frame, on top of which the receive arrays are mounted and later rotated jointly with the UE motion and the local self-rotation process.

2) *On-Device Array Placement*: For the concrete handset used in the MCMB-HDT, the platform carries $N_{\text{arr}} = 8$ receive arrays. In the local handset coordinate system, whose origin is taken at the handset center and whose coordinates are measured in centimeters, four FR3 arrays are placed near the four corners of the front face at approximately $(\pm 3.6, \pm 7.6, 0)$, while four FR1 arrays are placed at the centers of the four edges at approximately $(\pm 3.6, 0, 0)$ and $(0, \pm 7.6, 0)$. Their mounting locations and boresight orientations follow the 3GPP simulation assumptions in [7], as shown in Fig. 2. Because these arrays occupy different positions and orientations on the handset body, two arrays operating on the same carrier need

TABLE I
FEASIBLE-LINK INCIDENCE FOR THE FOUR-CELL, EIGHT-ARRAY
MCMB-HDT. A FILLED DOT INDICATES A FREQUENCY-MATCHED
CELL-ARRAY PAIR.

Cell	FR3 Corner Arrays				FR1 Edge Arrays			
	C1	C2	C3	C4	E1	E2	E3	E4
gNB TX0 FR1					•	•	•	•
gNB TX0 FR3	•	•	•	•				
gNB TX1 FR1					•	•	•	•
gNB TX1 FR3	•	•	•	•				

not observe the same effective link quality under the same external propagation field.

3) *Operating Bands and Radiation Patterns*: The handset configuration is intentionally multi-band. The FR1 arrays are modeled with dipole-like receive patterns at 3.5 GHz and 100 MHz bandwidth, while the FR3 arrays use more directional patch-like patterns at 15 GHz and 200 MHz bandwidth, as illustrated in Fig. 2. In the instantiated MCMB-HDT considered later, the handset interacts with $N_{\text{cell}} = 4$ cells carried by two gNB sites, and each site hosts one FR1 cell at 3.5 GHz and one FR3 cell at 15 GHz. Consistent with the later closed-loop formulation, each feasible link is determined by a frequency-matched cell-array pair (i, k) . For this four-cell, eight-array instantiation, each FR1 cell is connected to the four FR1 edge-center arrays and each FR3 cell is connected to the four FR3 corner arrays, yielding sixteen feasible links in total. Tab. I provides a compact visualization of this feasible-link incidence pattern. This band-dependent directivity is one of the main reasons why handset pose and self-rotation generate distinct CQI traces across the feasible links.

4) *FR1 and FR3 Radiation-Pattern Expressions*: To make the ray-tracing model fully reproducible, we next specify the receive radiation-pattern expressions used for the two frequency ranges. For the FR3 arrays at 15 GHz, we follow the handheld-UE pattern introduced in the 3GPP Release 19 model [7]. This pattern is defined through its vertical and horizontal cuts and then combined into a three-dimensional receive power pattern. In this notation, $\theta_{3\text{dB}}$ and $\phi_{3\text{dB}}$ denote the corresponding half-power beamwidths, while SLA_v denotes the vertical side-lobe attenuation limit. Specifically, the vertical cut is defined as

$$A''_{\text{dB}}(\theta'', \phi'' = 0^\circ) = -\min \left\{ 12 \left(\frac{\theta'' - 90^\circ}{\theta_{3\text{dB}}} \right)^2, \text{SLA}_v \right\}, \quad (1)$$

with $\theta_{3\text{dB}} = 125^\circ$, $\text{SLA}_v = 22.5$ dB, and $\theta'' \in [0^\circ, 180^\circ]$. The horizontal cut is given by

$$A'_{\text{dB}}(\theta' = 90^\circ, \phi') = -\min \left\{ 12 \left(\frac{\phi'}{\phi_{3\text{dB}}} \right)^2, A_{\text{max}} \right\}, \quad (2)$$

with $\phi_{3\text{dB}} = 125^\circ$, $A_{\text{max}} = 22.5$ dB, and $\phi' \in [-180^\circ, 180^\circ]$. The total FR3 radiation-power pattern is then written as

$$A_{\text{dB}}(\theta'', \phi'') = -\min \left\{ - \left(A''_{\text{dB}}(\theta'', \phi'' = 0^\circ) + A'_{\text{dB}}(\theta'' = 90^\circ, \phi'') \right), A_{\text{max}} \right\}. \quad (3)$$

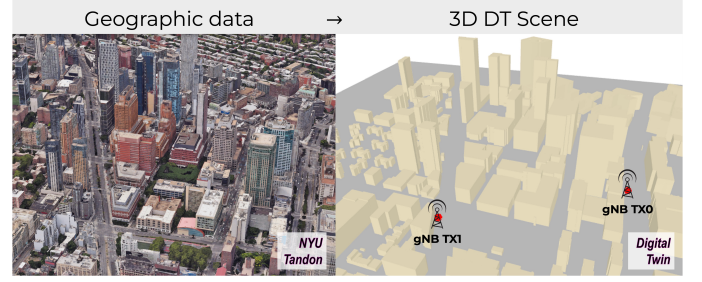


Fig. 3. From geographic data to the 3D DT scene used in the MCMB-HDT. The left panel shows the aerial view of the NYU Tandon School of Engineering campus area in Brooklyn, New York, and the right panel shows the corresponding Geo2SigMap-generated 3D DT scene imported into Sionna [17]. The two labeled sites, gNB TX0 and gNB TX1, denote the co-located FR1/FR3 gNB sites used in the ray-tracing simulations.

The corresponding linear power gain is

$$G_{\text{FR3}}(\theta'', \phi'') = 10^{A_{\text{dB}}(\theta'', \phi'')/10}. \quad (4)$$

For the FR1 arrays at 3.5 GHz, we use a dipole-type receive pattern. The implementation follows the dipole model provided in Sionna [18], which is based on the classical infinitesimal-dipole formulation [41]. Under the same local array-coordinate convention as above, the two field components are

$$C_\theta(\theta, \phi) = \sqrt{\frac{3}{2}} \sin \theta, \quad C_\phi(\theta, \phi) = 0. \quad (5)$$

The corresponding FR1 power-gain pattern is then

$$G_{\text{FR1}}(\theta, \phi) = |C_\theta(\theta, \phi)|^2 + |C_\phi(\theta, \phi)|^2 = \frac{3}{2} \sin^2 \theta. \quad (6)$$

5) *CQI and Rate Mapping*: For each feasible cell-array pair (i, k) generated by the MCMB-HDT, the ray-tracing engine produces the corresponding received power and propagation geometry over time. In this work, we summarize the instantaneous link quality by a CQI proxy γ_{ikt} , which is taken as the average received SNR on link (i, k) at time t . The achievable rate is then computed from γ_{ikt} via the practical CQI-rate mapping

$$R_{ikt} = \phi_i(\gamma_{ikt}) = B_i \min\{\rho_{\text{max}}, \beta \log_2(1 + \gamma_{ikt})\}, \quad (7)$$

where B_i is the bandwidth of cell i , β is the coding efficiency, and ρ_{max} is the practical spectral-efficiency cap [42], [43]. These CQI and rate traces form the basic DT outputs used later in the closed-loop array-activation problem.

B. Urban Environment and gNB Configuration

1) *Urban Building Model in Sionna*: The urban scene used in the MCMB-HDT is a geographically grounded outdoor model of the NYU Tandon School of Engineering campus area in Brooklyn, New York, which covers an approximately 1 km $\times$ 1 km region centered around 5 MetroTech Center. Rather than constructing a synthetic city layout, we generate the three-dimensional building model from geographic information using the Geo2SigMap pipeline in [17]. The resulting 3D meshes, including the building height coordinates associated with this region, can be directly imported into the Sionna

TABLE II
GEOGRAPHIC SCENE AND gNB-SITE SUMMARY FOR THE INSTANTIATED MCMB-HDT.

Parameter	Value
Scene area	NYU Tandon School of Engineering campus area, Brooklyn, NY
Reference address	5 MetroTech Center, Brooklyn, NY 11201
Geographic bounding box (latitude)	[40.69012764197041, 40.699120858029595]
Geographic bounding box (longitude)	[-73.99156687083165, -73.97970552916836]
gNB A site (lat, lon, alt.)	(40.694647, -73.988668, 10 m)
gNB A boresight target (lat, lon, alt.)	(40.695790, -73.988746, 1 m)
gNB B site (lat, lon, alt.)	(40.696316, -73.984515, 10 m)
gNB B boresight target (lat, lon, alt.)	(40.696284, -73.986282, 1 m)
Cells per site	One FR1 cell at 3.5 GHz / 100 MHz and one FR3 cell at 15 GHz / 200 MHz

ray-tracing engine, which allows the same scene geometry to be reused consistently across all gNB configurations, UE routes, and policy evaluations.

In this scene model, the dominant urban structures are represented by extruded building volumes together with a common ground plane, so that the ray tracer captures the main line-of-sight blockage, street-canyon visibility, and reflection structure induced by the surrounding city geometry. Following the official material library provided in Sionna, we assign *concrete* to the ground surface and *marble* to all building surfaces. These material assignments determine the reflection and penetration behavior used by the ray tracer and are kept fixed throughout all experiments. The correspondence between the real urban area and its DT instantiation in Sionna is illustrated in Fig. 3.

2) *gNB Locations and Cell Configuration*: The instantiated scene contains two gNB sites, denoted by gNB TX0 and gNB TX1 in Fig. 3. Their geographic locations, boresight targets, and the scene bounding box are summarized in Tab. II. At each site, we instantiate two co-located cells: one FR1 cell operating at 3.5 GHz with 100 MHz bandwidth and one FR3 cell operating at 15 GHz with 200 MHz bandwidth. The two co-located cells at each site share the same vertical-polarized gNB antenna configuration and the same boresight orientation. The resulting MCMB-HDT therefore contains four cells in total, two per gNB site, sharing the same urban scene and ray-tracing environment.

3) *Ray-Tracing Configuration*: The main ray-tracing and link-budget parameters used in the MCMB-HDT are summarized in Tab. III. The carrier frequencies and transmit powers are shared across all trajectories, while the CQI and rate traces are generated from the same Sionna propagation engine and the same urban scene described above.

C. UE Behavior Modeling in the DT

In this part of the MCMB-HDT, the handset is assumed to be carried by a pedestrian moving through the urban scene.

TABLE III
SIMULATION PARAMETERS USED IN THE MCMB-HDT.

Parameter	Value
Carrier frequency (bandwidth)	15 GHz (200 MHz) 3.5 GHz (100 MHz)
Tx antenna pattern	TR 38.901 vertical polarization
Transmit power per cell	51 dBm
Noise power spectral density	-174 dBm/Hz
Noise figure	7 dB
Measurement period	50 ms
Coding efficiency β	0.6
Max spectral efficiency $\rho_{\max}$	4.8 bits/Hz
Sionna ray types	specular, diffuse, refraction
Sionna max interaction depth	5 interactions

TABLE IV
MOBILITY-LEVEL PARAMETERIZATION FOR PEDESTRIAN MOTION AND HANDSET SELF-ROTATION.

Component	Parameter [unit]	Medium	High
Pedestrian	$\nu_{t+1} \in [\nu_{\min}, \nu_{\max}]$ [m/s]	[0, 5]	[0, 10]
	$\delta\psi_t^{\text{walk}} \in \mathcal{I}_{\text{dir}}$ [rad]	$[-\pi/6, \pi/6]$	$[-\pi/9, \pi/9]$
Handset	$\delta\psi_t^{\text{spin}} \in \mathcal{I}_{\psi}$ [rad]	$[-\pi/5, \pi/5]$	$[-\pi/5, \pi/5]$
	$\delta\vartheta_t^{\text{spin}} \in \mathcal{I}_{\vartheta}$ [rad]	$[-\pi/10, \pi/10]$	$[-\pi/10, \pi/10]$

Note: At each time step $\Delta t = 50$ ms, the pedestrian pair $(\nu_{t+1}, \delta\psi_t^{\text{walk}})$ is sampled uniformly from the Cartesian product of the listed speed and heading intervals, and the handset pair $(\delta\psi_t^{\text{spin}}, \delta\vartheta_t^{\text{spin}})$ is sampled uniformly from the Cartesian product of the listed yaw and pitch intervals. The sampled angular increments are then added to their current values.

We therefore separate the global motion of the pedestrian-carried handset from the local self-rotation of the handset body. The resulting UE behavior is modeled as a discrete-time stochastic dynamical system whose state evolves over the same measurement period used for channel generation.

Let $\Delta t = 50$ ms denote the discrete-time step. We describe the route evolution by the motion state

$$\mathbf{m}_t = [\mathbf{p}_t, \nu_t, \psi_t^{\text{walk}}, \psi_t^{\text{spin}}, \vartheta_t^{\text{spin}}], \quad (8)$$

where $\mathbf{p}_t \in \mathbb{R}^3$ denotes the UE position, ν_t is the walking speed, ψ_t^{walk} is the walking direction, and $(\psi_t^{\text{spin}}, \vartheta_t^{\text{spin}})$ are handset-orientation offsets introduced by local self-rotation.

1) *Pedestrian Route Generation*: At each time step, the pedestrian route is updated by jointly sampling the next-step walking speed and the walking-direction increment. Specifically,

$$\begin{bmatrix} \nu_{t+1} \\ \delta\psi_t^{\text{walk}} \end{bmatrix} \sim \text{Unif}([\nu_{\min}, \nu_{\max}] \times \mathcal{I}_{\text{dir}}), \quad (9a)$$

$$\psi_{t+1}^{\text{walk}} = \psi_t^{\text{walk}} + \delta\psi_t^{\text{walk}}, \quad (9b)$$

$$\mathbf{p}_{t+1} = \mathbf{p}_t + \nu_{t+1} \begin{bmatrix} \cos \psi_{t+1}^{\text{walk}} \\ \sin \psi_{t+1}^{\text{walk}} \\ 0 \end{bmatrix} \Delta t. \quad (9c)$$

Here, $\mathcal{I}_{\text{dir}}$ denotes the mobility-level-dependent sampling interval for the walking-direction increment. To ensure physically valid trajectories, the position is constrained to remain inside an admissible sampling region $\mathcal{R}_{\text{route}}$ while avoiding the building-occupied area $\mathcal{B}_{\text{bld}}$:

$$\mathbf{p}_t \in \mathcal{R}_{\text{route}}, \quad \mathbf{p}_t \notin \mathcal{B}_{\text{bld}}. \quad (10)$$

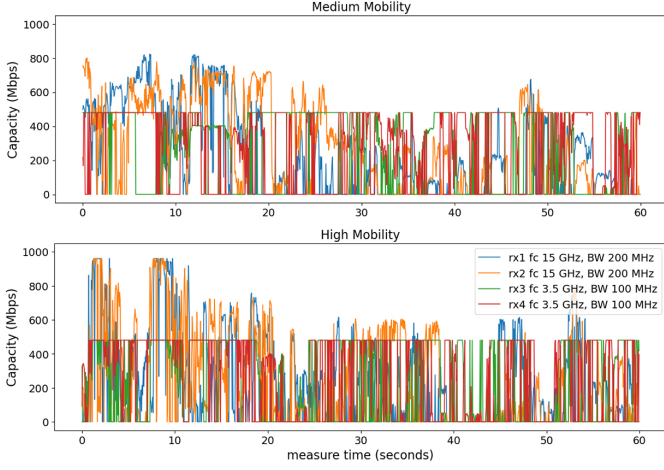


Fig. 4. Representative DT-generated capacity traces under medium and high UE mobility. Each panel shows the time-varying capacity observed by four representative RX arrays over a 60s trajectory, with RX0–RX1 operating at 15 GHz and RX2–RX3 operating at 3.5 GHz. The comparison illustrates how different mobility settings in the MCMC-HDT produce distinct temporal capacity dynamics across the enabled handset arrays.

Invalid transitions are rejected and resampled until the feasibility condition in (10) is satisfied.

2) *Handset Local Self-Rotation*: At each time step, the handset local self-rotation is updated by jointly sampling one yaw increment and one pitch increment:

$$\begin{bmatrix} \delta\psi_t^{\text{spin}} \\ \delta\vartheta_t^{\text{spin}} \end{bmatrix} \sim \text{Unif}(\mathcal{I}_\psi \times \mathcal{I}_\vartheta), \quad (11a)$$

$$\psi_{t+1}^{\text{spin}} = \psi_t^{\text{spin}} + \delta\psi_t^{\text{spin}}, \quad (11b)$$

$$\vartheta_{t+1}^{\text{spin}} = \vartheta_t^{\text{spin}} + \delta\vartheta_t^{\text{spin}}. \quad (11c)$$

The sampling intervals $\mathcal{I}_\psi$ and $\mathcal{I}_\vartheta$ depend on the selected mobility level and are summarized later in Tab. IV. This increment-based update captures natural variations in how the user rotates and tilts the handset while walking. Through (12), it directly affects the array boresight directions seen by the ray tracer and hence the time variation of the per-array CQI traces.

3) *Pedestrian Orientation and Global Handset Pose*: The global handset orientation is parameterized by ZYX Euler angles with roll fixed to zero. In this representation, the yaw angle ψ_t describes rotation around the global z -axis and the pitch angle ϑ_t describes rotation around the intermediate y -axis. The effective handset pose is obtained by combining the pedestrian walking direction with the local self-rotation:

$$\psi_t = \psi_t^{\text{walk}} + \psi_t^{\text{spin}}, \quad \vartheta_t = \vartheta_t^{\text{spin}}. \quad (12)$$

Hence, the walking direction ψ_t^{walk} controls the large-scale heading of the pedestrian-carried handset, while the spin terms account for how the handset is held relative to that heading.

4) *Mobility-Level Definitions*: Different mobility levels are obtained by changing the joint pedestrian-sampling support in (9a) and the joint handset self-rotation support in (11a). The concrete parameter settings used in this work are summarized in Tab. IV.

5) *DT-Generated Data Across Mobility Levels*: For each generated pedestrian trajectory and handset-pose realization,

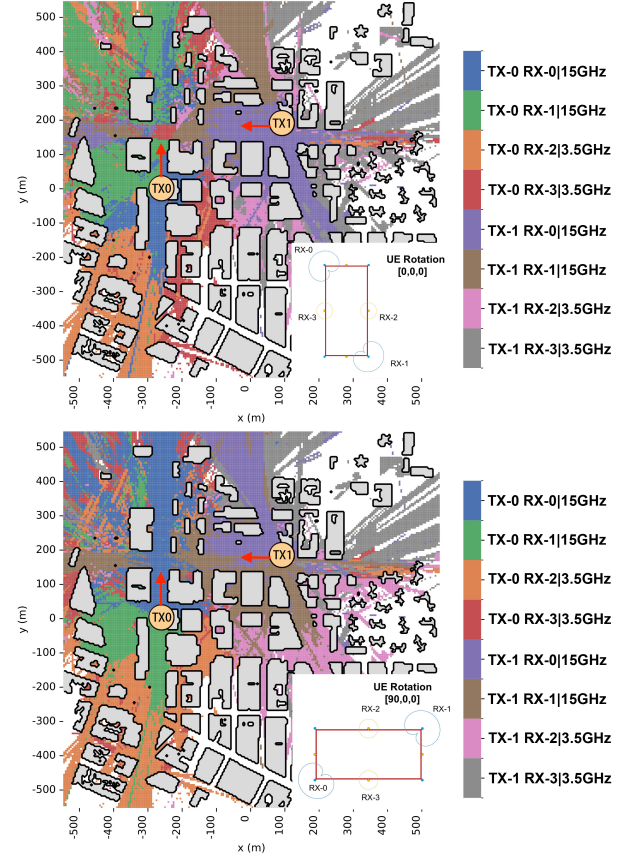


Fig. 5. Best-link maps generated by the MCMC-HDT with fixed global handset pose. The top and bottom panels fix the UE rotation to $[0^\circ, 0^\circ, 0^\circ]$ and $[90^\circ, 0^\circ, 0^\circ]$, respectively. For each sampled UE position, the color indicates the feasible gNB-cell-UE-array pair that achieves the largest capacity among the four enabled receive arrays, RX0–RX3. White regions indicate positions where none of the arrays receive a signal. The maps are shown with north upward and west to the left, and the inset in each panel illustrates the corresponding handset orientation and array directions.

the MCMC-HDT produces per-link CQI and rate sequences over time for the feasible cell–array pairs. To keep the qualitative visualization readable, Fig. 4 shows only the four representative receive arrays RX0–RX3, while the later closed-loop optimization uses the full eight-array handset configuration. The medium- and high-mobility examples produce different fluctuation rates and array rankings, showing that the RF observations available to the later controller depend directly on UE mobility and handset-pose evolution.

D. Ablation Study Design

The ablation study isolates three DT fidelity components: global handset pose, urban geometry, and handset-body modeling. First, Fig. 5 fixes the global handset pose and sweeps the UE position over the scene. Changing the global UE rotation changes which gNB-cell–array pair provides the largest capacity at many locations, demonstrating that handset orientation affects link quality spatially, not only along a single route.

Second, Fig. 6 compares the generated capacity traces under controlled changes to the environment and handset model. The column comparison isolates the effect of local UE self-rotation, while the row comparison isolates the effects of urban

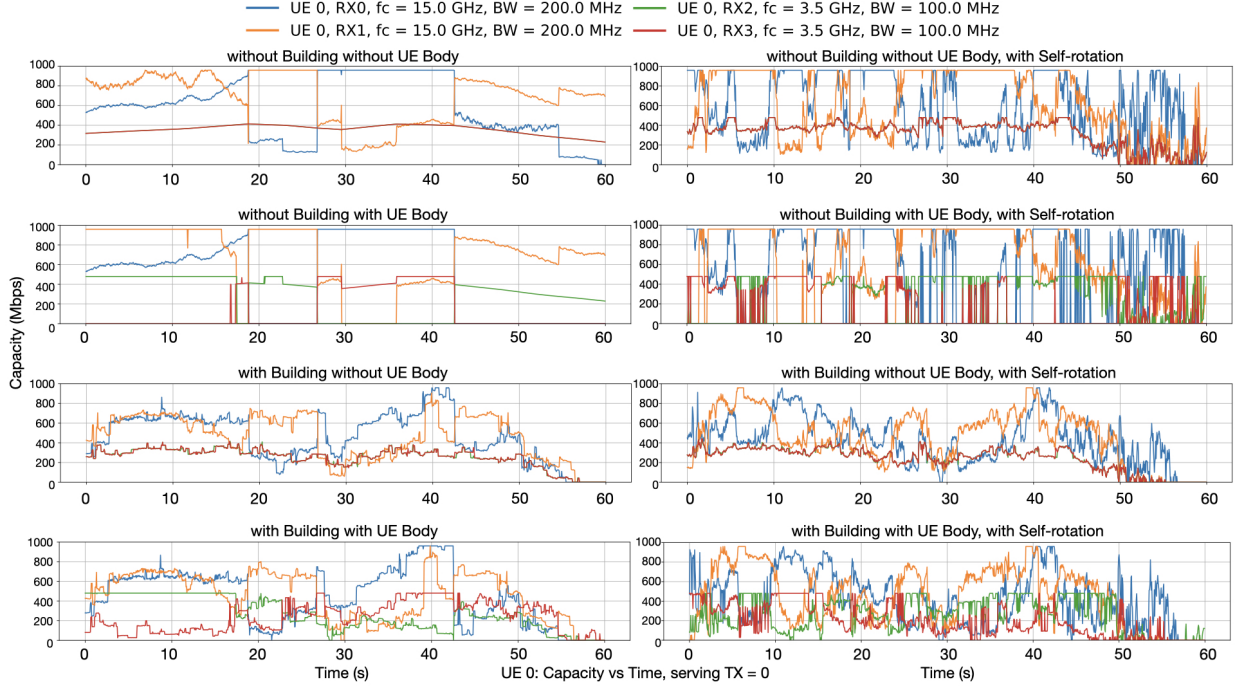


Fig. 6. Ablation study of the MCMB-HDT fidelity components. Each panel shows the DT-generated capacity traces over a 60s trajectory for the four representative receive arrays RX0–RX3, with the serving cell fixed to TX0 for visualization. The rows compare different environment and handset-modeling assumptions: without buildings and without the UE body, with buildings but without the UE body, without buildings but with the UE body, and the full environment with both buildings and the UE body. The left column uses a fixed handset local pose, while the right column adds UE self-rotation. The comparison shows how urban blockage, handset-body modeling, and local self-rotation jointly change the temporal structure and array dependence of the generated capacity traces.

spatial geometry and the UE handset body. The resulting traces differ in their rate levels, switching patterns across arrays, and temporal fluctuations, indicating that these modeling choices directly affect the data distribution used by the later control problem.

Connection to the Later Closed-Loop Control Problem.

The generated CQI and capacity traces form the data substrate for the measurement-limited array-activation problem in Section III. The later controller therefore operates on partial RF observations from a time-varying, pose-dependent link set rather than on fully observed static channels.

III. LEARNING-BASED CLOSED-LOOP ARRAY ACTIVATION

A. Problem Formulation

1) *System Model*: Let $\mathcal{F} \subset \mathbb{R}$ denote the set of carrier frequencies and let $\mathcal{X} \subset \mathbb{R}^3$ denote the spatial domain, where each location is represented by a 3D Cartesian coordinate. We represent a base station by its frequency–location pair $(f, x) \in \mathcal{F} \times \mathcal{X}$. Index the cells as $i = 1, \dots, N_{\text{cell}}$. Each cell is uniquely labeled by a frequency–location pair (f_i, x_i) , with $f_i \in \mathcal{F}$ and $x_i \in \mathcal{X}$. Similar to current deployments, multiple frequencies can be co-located at the base station location. We consider a single UE operating in an environment with N_{cell} cells. Let N_{arr} denote the number of UE antenna arrays, indexed by $k = 1, \dots, N_{\text{arr}}$. Each array k is associated with a carrier frequency f_k^{arr} . Different arrays on the same UE may operate on the same carrier frequency, for example, when they are located at different positions on the device. When active,

array k can receive transmissions from all cells whose carrier frequency equals f_k^{arr} , i.e., all cells i such that $f_i = f_k^{\text{arr}}$.

We define a channel, or equivalently a potential link, as the association between a cell and a UE array. Accordingly, the channel space is the set of all cell–array pairs

$$\mathcal{C} := \{(i, k) : i = 1, \dots, N_{\text{cell}}, k = 1, \dots, N_{\text{arr}}\}, \quad (13)$$

and the feasible channel space is restricted to frequency-matched pairs

$$\mathcal{C}_{\text{feas}} := \{(i, k) \in \mathcal{C} : f_i = f_k^{\text{arr}}\}. \quad (14)$$

For any $(i, k) \in \mathcal{C}$, let γ_{ikt} denote the link-quality metric for cell i measured on array k at time t , which we take as the (average) received SNR and use as a proxy for CQI. We set $\gamma_{ikt} = 0$ when $f_i \neq f_k^{\text{arr}}$, i.e., when the cell and array operate on different carrier frequencies. Following the DT measurement model in Section II-A5, we compute the achievable rate from γ_{ikt} via the practical mapping

$$R_{ikt} = \phi_i(\gamma_{ikt}) = B_i \min\{\rho_{\text{max}}, \beta \log_2(1 + \gamma_{ikt})\}. \quad (15)$$

where $\phi_i(\cdot)$ is the CQI–rate mapping function, B_i is the bandwidth of cell i , β is the coding efficiency, and ρ_{max} is the practical spectral-efficiency cap. In practice, the CQI can be measured from any periodic signal. For example, in 5G-NR, the CQI measurements can be performed from the SS block or CQI-RS [44].

Finally, even when only a subset of arrays is activated, transmissions from other cells operating on the same carrier can create co-channel interference. In this work, such interference

is treated as inherent in the measured γ and R data, and we do not separately model or estimate interference components.

2) *Action, Observation, and Serving Link*: In practice, the UE cannot measure all feasible links at every time step. This limitation comes from both hardware and operational considerations. On the hardware side, different arrays may share RF chains or require frequency-specific activation, so not all arrays can be measured simultaneously. On the operational side, activating multiple arrays at once may incur additional energy or processing cost. As a result, the closed-loop problem is inherently partially observed: at time t , the UE only observes the links associated with the selected active arrays.

We let A_t denote the set of active arrays in each time slot t . To conserve power, we consider a scenario where the UE can activate either one or two arrays in each time slot t . That is,

$$A_t = \{k_t^{\text{keep}}\} \text{ or } A_t = \{k_t^{\text{keep}}, a_t\}. \quad (16)$$

In (16), the $k_t^{\text{keep}} \in \{1, \dots, N_{\text{arr}}\}$ refers to an index of active arrays that has been maintained or “kept” from previous time slots. When $A_t = \{k_t^{\text{keep}}\}$, so there is only one active array, there is no exploration. In the case of two active arrays, $A_t = \{k_t^{\text{keep}}, a_t\}$, the index a_t represents a second array that is activated for the purpose of exploring potential new links.

Given the selected subset A_t , the UE observes all feasible channels associated with the active arrays:

$$\mathcal{O}_t := \{(i, k) \in \mathcal{C}_{\text{feas}} : k \in A_t\}, \quad (17)$$

and

$$o_t = \{\hat{\gamma}_{ikt} : (i, k) \in \mathcal{O}_t\}. \quad (18)$$

Among the observed links, the serving link is selected as

$$c_t = (i_t^*, k_t^*) \in \arg \max_{(i, k) \in \mathcal{O}_t} \hat{\gamma}_{ikt}. \quad (19)$$

The retained array used at the next step is then updated to the array index of the selected serving link:

$$k_{t+1}^{\text{keep}} = k_t^*. \quad (20)$$

Hence, whenever exploration reveals a better link on the newly tested array, that array becomes the retained array at the following step.

Any mechanism can be used for selecting the serving cell. The model is consistent with carrier aggregation [45], where the UE has active connections with multiple cell sites simultaneously. Once the CQI across different cells are measured and reported back to the network, the network will tend to automatically schedule the UE on the optimal link, assuming the loading is not an issue. However, the formulation is general, and the link selection could also be implemented with a handover, although the delay would be longer.

3) *Risk State and Reward Design*: For the policies that use risk-state control, we introduce a dynamic risk state z_t to regulate the long-term usage of two-array actions. We initialize $z_1 = 1$ and update it according to

$$z_{t+1} = (1 - \alpha)z_t + \alpha|A_t|, \quad 0 < \alpha \leq 1. \quad (21)$$

Let the threshold be written as

$$b = 1 + \varepsilon, \quad 0 < \varepsilon < 1. \quad (22)$$

Whenever $z_t \geq b$, the policy is forced to choose

$$a_t = k_t^{\text{keep}}, \quad (23)$$

so exploration is disabled and only the retained array remains active. Throughout this paper, we use ε as a unified *exploration-control* parameter across the five algorithms considered later. For the ε -greedy benchmark, ε is the fixed per-step probability of testing one additional array. For *PPO-rate*, *PPO-risk*, *bandit-rate*, and *bandit-risk*, ε enters through the threshold $b = 1 + \varepsilon$ and therefore controls exploration indirectly through the long-term dynamics of z_t and the rule (23).

We consider two reward designs. The rate-maximization reward is

$$r_t^{\text{rate}} = \log R_{c_t}. \quad (24)$$

The risk-aware reward is

$$r_t^{\text{risk}} = \log R_{c_t} \left[e^{-\lambda_1(z_t-1) \frac{R_{c_t}}{\lambda_2 + R_{c_t}}} \right], \quad (25)$$

where $\lambda_1 > 0$ controls the overall strength of the risk-aware penalty and $\lambda_2 > 0$ determines how quickly the penalty becomes active as the achieved rate increases. With this design, the effect of the risk state is weak when the achieved rate is small, and becomes stronger when the achieved rate is large. Hence, the policy is allowed to tolerate a higher risk level when the current communication quality is poor, while being encouraged to keep the risk level low when the current rate is already high.

4) *Interpretation of the Risk Dynamics and Empirical Exploration Rate*: The following interpretation applies to *PPO-rate*, *PPO-risk*, *bandit-rate*, and *bandit-risk*, all of which use the risk state z_t . Define the exploration indicator

$$e_t = \mathbb{1}\{a_t \neq k_t^{\text{keep}}\}, \quad (26)$$

so that $|A_t| = 1 + e_t$. Thus, the shifted variable $y_t = z_t - 1$ satisfies

$$y_{t+1} = (1 - \alpha)y_t + \alpha e_t, \quad (27)$$

so y_t is an exponentially weighted moving average of the exploration indicator. Accordingly, α controls the effective memory of the risk state, which is on the order of $1/\alpha$. That is

$$y_t = (1 - \alpha)^t y_0 + \alpha \sum_{\tau=0}^{t-1} (1 - \alpha)^{t-1-\tau} e_\tau. \quad (28)$$

The half-life, namely the number of steps required for the influence of past exploration to decay by one half, is

$$T_{1/2} = \frac{\ln(1/2)}{\ln(1 - \alpha)} \approx \frac{0.693}{\alpha}, \quad (29)$$

where the approximation becomes more accurate for smaller values of α .

Moreover, in a stationary regime, the distribution of y_t does not change with t , so $\mathbb{E}[y_{t+1}] = \mathbb{E}[y_t]$. Taking expectations in (27), then gives

$$\mathbb{E}[y_t] = (1 - \alpha)\mathbb{E}[y_t] + \alpha\mathbb{E}[e_t], \quad (30)$$

which implies

$$\mathbb{E}[y_t] = \mathbb{E}[e_t]. \quad (31)$$

Thus, the long-term fraction of two-array actions is tracked by the average level of y_t , or equivalently by the average level of z_t after subtracting one. Hence, with $b = 1 + \varepsilon$, the parameter ε provides a rough control of the long-term exploration rate for *PPO-rate*, *PPO-risk*, *bandit-rate*, and *bandit-risk*. In contrast, for ε -greedy, the exploration rate is directly fixed to ε at every step by construction.

For a unified experimental comparison, we also define a post-hoc empirical exploration rate. Given one rollout of length H , let the sequence $\{z_t\}$ be either the internal risk state used by the policy or the same recursion in (21) applied offline to the realized action sequence when the policy itself does not maintain a risk state, as in ε -greedy. Define the empirical average risk level as

$$\hat{Z} := \frac{1}{H} \sum_{t=1}^H z_t, \quad (32)$$

and, since $y_t = z_t - 1$, define the empirical average of the shifted variable as

$$\hat{Y} := \frac{1}{H} \sum_{t=1}^H y_t = \frac{1}{H} \sum_{t=1}^H (z_t - 1). \quad (33)$$

We then define the empirical exploration rate as

$$\hat{\varepsilon} := \mathbb{E}[\hat{Y}]. \quad (34)$$

Equivalently,

$$\hat{\varepsilon} = \mathbb{E}[\hat{Z}] - 1. \quad (35)$$

In practice, the expectation is estimated by averaging $\hat{Y}$, or equivalently $\hat{Z} - 1$, over the evaluation trajectories.

B. PPO-Based Closed-Loop Array Activation

The closed-loop decision problem above is partially observed since the UE measures only the channels associated with the active arrays. To learn a closed-loop policy, we use proximal policy optimization (PPO) [19], an on-policy policy-gradient method. To summarize the observation history, PPO maintains a recurrent hidden state h_t . When $z_t < b$, PPO defines a distribution over array actions

$$\pi_\theta(a \mid h_t, k_t^{\text{keep}}, z_t), \quad a \in \{1, \dots, N_{\text{arr}}\}, \quad (36)$$

and samples

$$a_t \sim \pi_\theta(\cdot \mid h_t, k_t^{\text{keep}}, z_t). \quad (37)$$

When $z_t \geq b$, PPO is constrained by (23) and sets $a_t = k_t^{\text{keep}}$. Together, these two cases define the overall PPO policy, which we still denote by π_θ . Thus, selecting $a_t = k_t^{\text{keep}}$ means that PPO keeps the retained array and does not explore, while

selecting $a_t \neq k_t^{\text{keep}}$ means that PPO explores array a_t in addition to the retained array. The active-array set A_t is then determined by (16).

After executing the action, the recurrent hidden state is updated as

$$h_{t+1} = f_\theta(h_t, o_t, a_t, \tilde{r}_t), \quad (38)$$

where $f_\theta(\cdot)$ is implemented by a recurrent network and $\tilde{r}_t$ denotes the reward used by the selected PPO variant.

We consider two PPO variants:

$$J^{\text{rate}}(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=1}^H r_t^{\text{rate}} \right], \quad (39)$$

which corresponds to the rate-maximization reward in (24), and

$$J^{\text{risk}}(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=1}^H r_t^{\text{risk}} \right], \quad (40)$$

which corresponds to the risk-aware reward in (25). In the experiments, these two variants are referred to as *PPO-rate* and *PPO-risk*, respectively.

C. Benchmark Policies

1) ε -Greedy: We first consider an ε -greedy benchmark that explicitly explores by testing one new array against the currently retained array. Let $k_t^{\text{keep}} \in \{1, \dots, N_{\text{arr}}\}$ denote the retained array at the beginning of step t , and initialize k_1^{keep} arbitrarily. With probability $1 - \varepsilon$, the benchmark keeps the retained array and sets

$$a_t = k_t^{\text{keep}}. \quad (41)$$

With probability ε , it explores by sampling one additional array uniformly from the remaining arrays:

$$a_t \sim \text{Unif}(\{1, \dots, N_{\text{arr}}\} \setminus \{k_t^{\text{keep}}\}). \quad (42)$$

The active-array set A_t is again determined by (16). After executing A_t , the UE forms $\mathcal{O}_t$, selects the serving link, and updates the retained array via (20). This benchmark does not maintain a risk state internally, and its per-step exploration probability is exactly ε by construction.

2) *Bandit Policies*: We also consider two bandit benchmarks, in the spirit of standard action-value methods for bandit problems [46], that follow the same retained-array semantics and the same risk-state control mechanism as PPO, but replace the recurrent policy with a softmax bandit over arrays. For each variant $\nu \in \{\text{rate}, \text{risk}\}$, the policy maintains a logit

$$v_t^\nu(k), \quad k = 1, \dots, N_{\text{arr}}, \quad (43)$$

for each array. These logits define the array-selection distribution

$$\pi_t^{\text{band}, \nu}(k) = \frac{e^{v_t^\nu(k)/\eta}}{\sum_{k'=1}^{N_{\text{arr}}} e^{v_t^\nu(k')/\eta}}, \quad k = 1, \dots, N_{\text{arr}}, \quad (44)$$

where $\eta > 0$ is the temperature parameter. When $z_t < b$, the executed array action is sampled as

$$a_t \sim \pi_t^{\text{band}, \nu}(\cdot). \quad (45)$$

When $z_t \geq b$, the benchmark follows (23) and sets $a_t = k_t^{\text{keep}}$. Hence, selecting $a_t = k_t^{\text{keep}}$ means no exploration, while selecting $a_t \neq k_t^{\text{keep}}$ means that array a_t is explored in addition to the retained array. The active-array set A_t is then formed according to (16). After executing A_t , we define the best observed rate on each active array by

$$\bar{R}_t(k) := \max_{i:(i,k) \in \mathcal{O}_t} \phi_i(\hat{\gamma}_{ikt}), \quad k \in A_t. \quad (46)$$

The corresponding array-wise rate reward is

$$\bar{r}_t^{\text{rate}}(k) = \log \bar{R}_t(k), \quad k \in A_t. \quad (47)$$

This quantity is the local counterpart of (24), while the global reward satisfies

$$r_t^{\text{rate}} = \max_{k \in A_t} \bar{r}_t^{\text{rate}}(k). \quad (48)$$

Similarly, by replacing the achieved rate in (25) with $\bar{R}_t(k)$, we define the corresponding array-wise risk-aware reward as

$$\bar{r}_t^{\text{risk}}(k) = \log \bar{R}_t(k) e^{-\lambda_1(z_t-1) \frac{\bar{R}_t(k)}{\lambda_2 + \bar{R}_t(k)}}, \quad k \in A_t. \quad (49)$$

The logits are then updated according to

$$v_{t+1}^\nu(k) = \begin{cases} (1 - \kappa)v_t^\nu(k) + \kappa \bar{r}_t^\nu(k), & k \in A_t, \\ v_t^\nu(k), & k \notin A_t, \end{cases} \quad (50)$$

while the retained array and the risk state evolve according to (20) and (21), respectively. The two resulting variants are referred to as *bandit-rate* for $\nu = \text{rate}$ and *bandit-risk* for $\nu = \text{risk}$. Hence, if the benchmark does not explore, only the retained array is updated. If it explores, both active arrays are updated using the best observed link on each array.

Summary of policies: The experiments below compare five policies. The *PPO-rate* and *PPO-risk* policies share the same single-stage recurrent policy architecture with inputs $(h_t, k_t^{\text{keep}}, z_t)$ and the same retained-array transition, and differ only in whether they optimize the rate reward in (24) or the risk-aware reward in (25). The *ϵ -greedy* policy serves as a heuristic benchmark with explicit per-step exploration probability ϵ . The *bandit-rate* and *bandit-risk* policies use the same array-level softmax bandit construction and the same retained-array and risk-state transitions, and differ only in whether their logits are updated by the local rate reward $\bar{r}_t^{\text{rate}}(k)$ or the local risk-aware reward $\bar{r}_t^{\text{risk}}(k)$ in (50).

IV. PERFORMANCE EVALUATION OF CLOSED-LOOP ARRAY ACTIVATION

A. Experimental Setting

1) *Dataset and Evaluation Scene:* The experiments are conducted on the NYU Tandon Campus scene shown in Fig. 7. The study area is divided into five districts, D1–D5. We use D1 and D3–D5 as the training districts and reserve D2 as the held-out test district. This split evaluates whether a policy trained on one set of campus regions can operate on trajectories generated in an unseen region.

In each training district, we generate 500 medium-mobility and 500 high-mobility trajectories, giving 4000 training trajectories across D1 and D3–D5. The held-out district D2 also contains 500 medium-mobility and 500 high-mobility trajectories

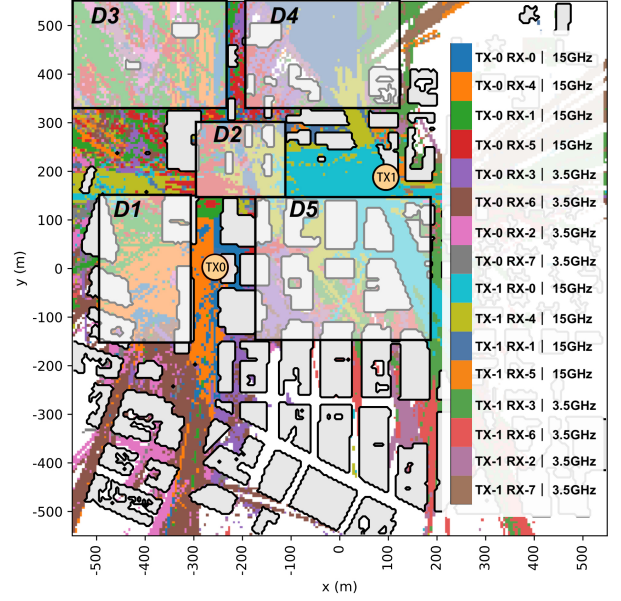


Fig. 7. Dataset distribution across the study area, the color indicates the feasible gNB-cell-UE-array pair that achieves the largest capacity among all eight arrays. The region is divided into five districts (D1–D5) based on the NYU Tandon Campus. UE trajectories are sampled within each district. District D2 is reserved as the test set, while the remaining four districts are used for training. Each UE is equipped with 8 RX arrays, and communication is established with 2 serving cells from two TXs. The overlaid best-link map indicates different regional difficulty levels: D1 and D4 are relatively link-stable, D5 is moderately affected by edge and blockage conditions, and D2/D3 show more mixed link dominance due to stronger geometry-induced variation.

for testing. Each trajectory lasts 60 s with a 0.05 s sampling interval, yielding 1200 time samples per trajectory. The closed-loop evaluation uses the eight-array UE configuration and two serving TX cells from the MCMB-HDT.

2) *PPO Training Setup:* The PPO-rate and PPO-risk policies are trained with the reward and rollout settings in Tab. V. The RF CQI and rate traces are pre-generated by the MCMB-HDT. During PPO training, each trajectory environment replays one sampled route trace and exposes only the measurements associated with the arrays activated by the current policy, so the resulting rollout samples are on-policy interaction samples rather than newly ray-traced RF samples. Configurations with $\lambda_1 > 0$ use the risk-aware reward in (25). When $\lambda_1 = 0$, the exponential penalty in (25) becomes one, so the reward reduces to the rate reward in (24). In this case, λ_2 has no effect and only one λ_2 value is evaluated.

3) *Benchmark Policies:* We compare PPO against the benchmark policies defined in Section III: bandit-rate, bandit-risk, and ϵ -greedy. The benchmark parameters are summarized in Tab. VI. All policies select the serving link as the best observed RX–TX link among the active arrays and are evaluated on the same D2 test trajectories.

4) *Exploration Rate and Performance Metric:* The parameter ϵ has different operational meanings across policies. For ϵ -greedy, it is the explicit per-step probability of testing one additional array, so the reported exploration rate is the design value ϵ . For PPO-rate, PPO-risk, bandit-rate, and bandit-risk, ϵ sets the risk threshold $b = 1 + \epsilon$ and therefore acts as a target exploration level through the risk state z_t . As shown in Section III, the stationary risk-state relation suggests that

TABLE V
PPO TRAINING CONFIGURATION.

Parameter	Value
Risk-control sweep	$\varepsilon \in \{0.3, 0.5, 0.7, 0.8, 0.9\}$
Risk penalty sweep	$\lambda_1 \in \{0, 0.1, 0.3, 0.5, 0.8\}$
Rate-scaling sweep	$\lambda_2 \in \{300, 600\}$, with one value when $\lambda_1 = 0$
Actor and critic networks	Separate single-layer GRUs, hidden size 256
Learning rates	Actor 1×10^{-4} and critic 5×10^{-5}
On-policy step data per PPO rollout	512 parallel trajectory environments $\times$ 256 decision steps = 131,072 transitions
Training length	2000 PPO rollout per run
Hardware and runtime	NVIDIA RTX A6000, 3.5 GPU hours per PPO model

TABLE VI
BENCHMARK POLICY CONFIGURATION.

Policy	Configuration
bandit-rate	Retained-array softmax bandit with $\eta = 1.0$ and $\kappa = 0.03$, rate reward, and $\varepsilon \in \{0.3, 0.5, 0.8\}$
bandit-risk	Retained-array softmax bandit with $\eta = 1.0$ and $\kappa = 0.03$, risk-aware reward, and the same ε , λ_1 , and λ_2 sweeps as PPO
ε -greedy	Explicit exploration probability $\varepsilon \in \{0.1 : 0.1 : 0.9\}$, with one additional array sampled uniformly during exploration

the long-term two-array action fraction should be close to $b - 1 = \varepsilon$. In finite evaluations, however, the realized action distribution also depends on the reward parameters (λ_1, λ_2) and on whether the action probabilities are produced by PPO or by the bandit update. We therefore report the empirical exploration rate $\hat{\varepsilon} = \mathbb{E}[\hat{Z}] - 1$ defined in (35). With a slight abuse of notation, Section IV-C uses $\hat{\varepsilon}$ as the common reported exploration-rate axis: for ε -greedy this value is the design parameter ε , while for PPO and bandit policies, it is the empirical estimate from (35).

To compare policies at different reported exploration rates, we use the route-level best-rate ratio. Let π denote any evaluated policy, and let R_t^π be the achievable rate selected by π at step t . This is the physical rate in (15), not the logarithmic reward in (24) or (25). For a route of length H , we define

$$\text{BR}^\pi = \frac{\sum_{t=1}^H R_t^\pi}{\sum_{t=1}^H \max_{(i,k) \in \mathcal{C}_{\text{feas}}} R_{ikt}}. \quad (51)$$

Thus, BR^π is the ratio between the route-level cumulative rate achieved by policy π and the route-level cumulative oracle rate obtained by selecting the best feasible link at each step.

B. Trajectory-Level Policy Behavior

We first inspect one representative held-out trajectory to illustrate how the risk-aware mechanism affects sequential array-activation decisions. The comparison in Fig. 8 fixes $\varepsilon = 0.3$, $\lambda_1 = 0.3$, and $\lambda_2 = 300$, and includes PPO-risk, bandit-risk, ε -greedy, and the oracle that selects the

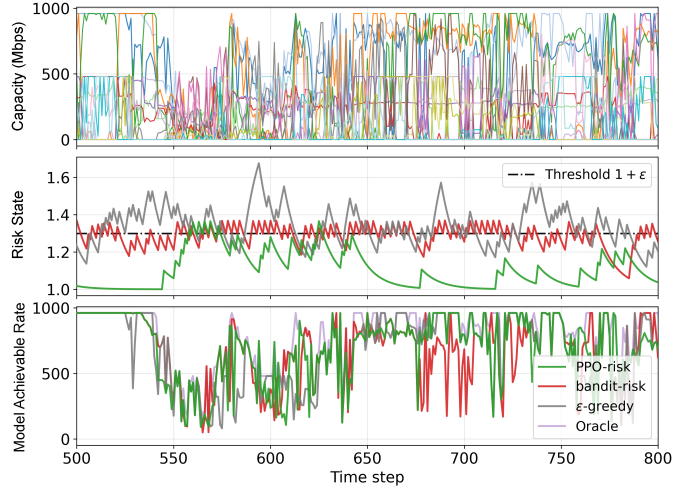


Fig. 8. Trajectory-level comparison of risk-aware closed-loop array activation for a representative held-out route with $\varepsilon = 0.3$, $\lambda_1 = 0.3$, and $\lambda_2 = 300$. The top panel shows the instantaneous capacity traces of the feasible links generated by the MCMC-HDT. The middle panel shows the risk state z_t and the threshold $1 + \varepsilon$. The bottom panel shows the physical achievable rate selected by PPO-risk, bandit-risk, ε -greedy, and the oracle best-link selection. PPO-risk keeps the risk state mostly below the threshold while tracking high-rate opportunities more effectively than the benchmark policies.

TABLE VII
FIXED-SETTING ROUTE-LEVEL PERFORMANCE FOR $\varepsilon = 0.3$, $\lambda_1 = 0.3$, AND $\lambda_2 = 300$. MEAN METRICS ARE REPORTED AS MEAN $\pm$ STD, AND BR COVERAGE ENTRIES ARE PERCENTAGES OF HELD-OUT ROUTES.

Metric	PPO-risk	bandit-risk	ε -greedy
BR^π	0.867 $\pm$ 0.065	0.843 $\pm$ 0.076	0.817 $\pm$ 0.080
Rate (Mbps)	740.2 $\pm$ 127.7	719.9 $\pm$ 129.7	698.9 $\pm$ 131.2
Risk reward	6.339 $\pm$ 0.297	6.103 $\pm$ 0.243	5.973 $\pm$ 0.245
Mean risk state	1.117 $\pm$ 0.059	1.251 $\pm$ 0.061	1.297 $\pm$ 0.013
$\text{BR}^\pi \geq 0.90$	31.8%	23.8%	22.6%
$\text{BR}^\pi \geq 0.85$	56.8%	44.4%	31.8%
$\text{BR}^\pi \geq 0.80$	83.8%	70.2%	53.4%
$\text{BR}^\pi \geq 0.75$	96.6%	89.0%	77.0%

best feasible link at each step. The rate-oriented variants are omitted from this trajectory-level view because the goal here is to isolate the temporal effect of risk-aware decision making. The full rate model and statistical comparison are reported in Section IV-C.

The top panel shows that the feasible-link capacities vary unevenly along the trajectory: in some intervals the dominant high-rate links change rapidly, while in other intervals several arrays provide similarly strong rates or the best array remains relatively stable. The middle and bottom panels show that bandit-risk and ε -greedy can maintain a high risk state while still selecting lower-rate links. In contrast, PPO-risk keeps the risk state under control, with z_t remaining mostly below the threshold $1 + \varepsilon$ and returning toward the low-risk baseline after short exploration bursts, while maintaining a selected rate closer to the oracle. When the capacity traces indicate that the currently observed information is sufficient, such as when the dominant array is stable or multiple arrays offer high capacity, PPO-risk tends to reduce unnecessary exploration. This behavior shows that the recurrent policy is not simply using exploration at a fixed rate, but adapting array activation to the temporal structure of the DT-generated RF observations.

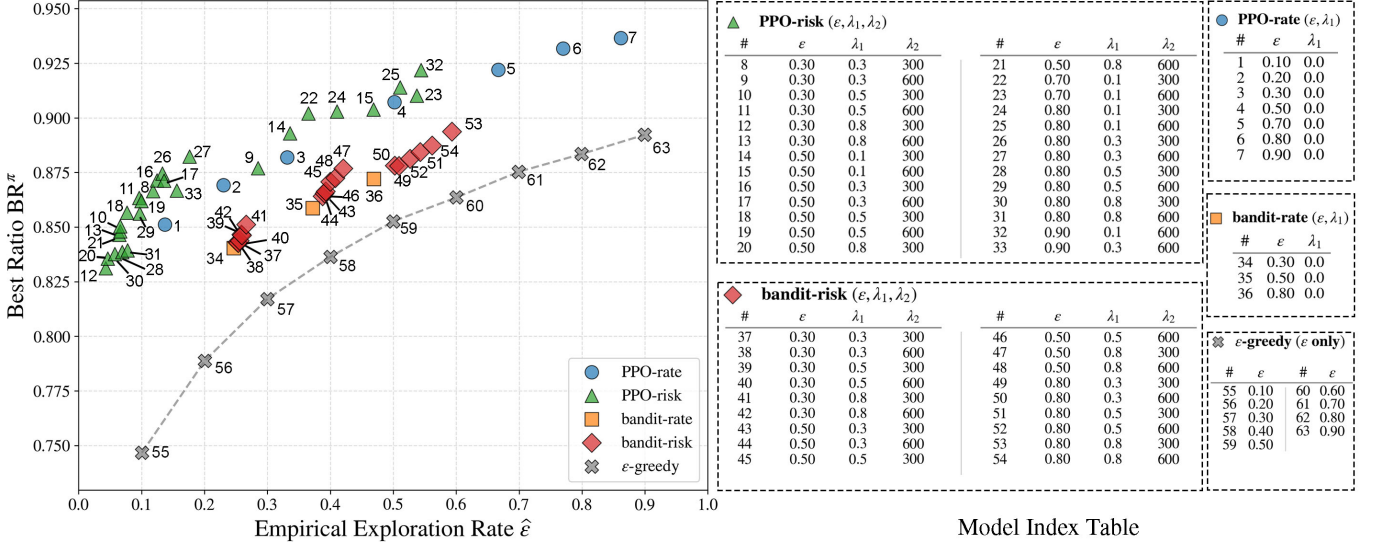


Fig. 9. Statistical comparison of route-level best-rate ratio and empirical exploration rate over the held-out test routes. The left panel plots each evaluated configuration using the empirical exploration rate $\hat{\epsilon}$ from (35) on the horizontal axis and the best-rate ratio BR^π from (51) on the vertical axis. Marker styles indicate PPO-rate, PPO-risk, bandit-rate, bandit-risk, and ϵ -greedy. The right panel maps each indexed point in the scatter plot to its hyperparameter setting $(\epsilon, \lambda_1, \lambda_2)$, with λ_2 omitted for rate-only settings where the risk penalty is disabled.

Tab. VII shows that PPO-risk achieves the best rate, risk reward, and route-level BR coverage while maintaining the lowest average risk state, indicating better use of the same nominal risk-control setting.

C. Statistical Performance Comparison

Fig. 9 compares all evaluated policies across the hyperparameter sweeps. The ϵ -greedy curve forms a clear lower envelope: increasing explicit exploration raises the reported $\hat{\epsilon}$, but the random additional-array selection does not translate efficiently into higher BR^π . The bandit policies improve over ϵ -greedy because their action probabilities are adapted from observed rewards, but their points remain below the PPO policies and the risk-aware bandit does not substantially reduce empirical exploration.

The PPO policies achieve the strongest trade-off between route-level rate performance and empirical exploration. PPO-rate attains the highest BR^π values because it optimizes the rate reward without the risk penalty. PPO-risk slightly reduces BR^π relative to PPO-rate, but it shifts the operating points to much lower $\hat{\epsilon}$, showing that the risk-aware reward can reduce unnecessary two-array activation while preserving most of the route-level achievable-rate gain. This confirms the role of the proposed risk-aware closed-loop policy: it does not merely increase exploration to find better links, but learns when exploration is informative enough to justify its cost.

V. CONCLUSION

This paper presented MCMB-HDT, a UE-centric multi-cell multi-band handset digital twin for closed-loop wireless control. Unlike network-only DT workflows, the proposed framework jointly models the urban propagation environment, multi-cell multi-band infrastructure, handset-mounted antenna

arrays, UE body effects, and stochastic mobility and pose dynamics. The resulting DT-generated RF traces provide a controlled substrate for studying measurement-limited array activation under realistic handset and environment interactions.

Through DT visualization and ablation studies, we showed that UE mobility, global handset pose, local self-rotation, building geometry, and the handset body model all materially affect the observed capacity traces and best-link structure. These results indicate that UE-side geometry and behavior are not secondary details, but core factors shaping the data distribution seen by closed-loop wireless controllers.

Building on this DT substrate, we formulated a closed-loop array-activation problem and evaluated PPO, bandit, and ϵ -greedy policies under partial RF observation. The results show that PPO-based policies achieve stronger best-rate-ratio performance than bandit and ϵ -greedy baselines. In particular, PPO-risk preserves most of the route-level rate gain while reducing empirical exploration rate, demonstrating an effective trade-off between link discovery and measurement overhead. These findings support UE-centric handset DTs as practical twin-in-the-loop tools for learning-based, resource-aware wireless adaptation in future digital twin networks.

REFERENCES

- [1] S. Kang, M. Mezzavilla, S. Rangan, A. Madanayake, S. B. Venkatakrishnan, G. Hellbourn, M. Ghosh, H. Rahmani, and A. Dhananjay, "Cellular wireless networks in the upper mid-band," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 2058–2075, 2024.
- [2] P. Testolina, M. Polese, and T. Melodia, "Sharing spectrum and services in the 7–24 ghz upper midband," *IEEE Communications Magazine*, vol. 62, no. 8, pp. 170–177, 2024.
- [3] D. Shakyia, M. Ying, T. S. Rappaport, H. Poddar, P. Ma, Y. Wang, and I. Al-Wazani, "Propagation measurements and channel models in Indoor Environment at 6.75 GHz FR1 (C) and 16.95 GHz FR3 Upper-mid band Spectrum for 5G and 6G," in *GLOBECOM 2024-2024 IEEE Global Communications Conference*. IEEE, 2024, pp. 998–1003.

- [4] 3GPP, "TSG RAN NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone (Release 18)," 3GPP, Tech. Rep. TS 38.101-1, June 2024, version 18.3.0.
- [5] Y. Wang, L. Sun, Z. Du, and Z. Zhang, "Review antenna design for modern mobile phones: A review," *Electromagnetic Science*, vol. 2, no. 2, pp. 1–36, 2024.
- [6] S. Svendsen, D. Gold, C. Rom, V. Pauli, and V. Nurmela, "Realistic UE Antennas for 6G in the 3GPP Channel Model," *IEEE Communications Standards Magazine*, vol. 10, no. 1, pp. 109–115, 2026.
- [7] 3GPP, "TSG RAN WG1#121; Introduction of Rel-19 7-24 GHz channel model enhancements," 3GPP, Tech. Rep. R1-2504964, May 2025, r38901, CR0026 version 18.0.0.
- [8] S. Svendsen, D. Gold, C. Rom, V. Pauli, and V. Nurmela, "Realistic UE antennas for 6G in the 3GPP channel model," *IEEE Commun. Standards Mag.*, vol. 10, no. 1, pp. 109–115, 2026.
- [9] V. Raghavan, O. Koymen, and J. Li, "Discrete vs. modular antenna tradeoffs for upper mid-bands," in *Proc. 2025 59th Asilomar Conf. Signals, Syst., Comput.*, 2025, pp. 824–830.
- [10] V. Raghavan, S. Noimanivone, S. K. Rho, B. Farin, P. Connor, R. A. Motos, Y.-C. Ou, K. Ravid, M. A. Tassoudji, O. H. Koymen *et al.*, "Hand and body blockage measurements with form-factor user equipment at 28 GHz," *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 1, pp. 607–620, 2021.
- [11] P. Liu, I. Syrytsin, J. Ø. Nielsen, G. F. Pedersen, and S. Zhang, "Characterization and modeling of the user blockage for 5G handset antennas," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–11, 2020.
- [12] D. Shakya, M. Ying, T. S. Rappaport, H. Poddar, P. Ma, Y. Wang, and I. Al-Wazani, "Comprehensive FR1 (C) and FR3 lower and upper mid-band propagation and material penetration loss measurements and channel models in indoor environment for 5G and 6G," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 5192–5218, 2024.
- [13] C. Slezak, V. Semkin, S. Andreev, Y. Koucheryavy, and S. Rangan, "Empirical effects of dynamic human-body blockage in 60 GHz communications," *IEEE Communications Magazine*, vol. 56, no. 12, pp. 60–66, 2018.
- [14] G. R. MacCartney, T. S. Rappaport, and S. Rangan, "Rapid fading due to human blockage in pedestrian crowds at 5G millimeter-wave frequencies," in *GLOBECOM 2017-2017 IEEE Global Communications Conference*. IEEE, 2017, pp. 1–7.
- [15] M. Polese, R. Jana, and M. Zorzi, "TCP and MP-TCP in 5G mmWave networks," *IEEE Internet Computing*, vol. 21, no. 5, pp. 12–19, 2017.
- [16] M. Zhang, M. Polese, M. Mezzavilla, J. Zhu, S. Rangan, S. Panwar, and M. Zorzi, "Will TCP work in mmWave 5G cellular networks?" *IEEE Communications Magazine*, vol. 57, no. 1, pp. 65–71, 2019.
- [17] Y. Li, Z. Li, Z. Gao, and T. Chen, "Geo2sigmap: High-fidelity rf signal mapping using geographic databases," in *Proc. 2024 IEEE Int. Symp. Dyn. Spectrum Access Netw. (DySPAN)*, 2024, pp. 277–285.
- [18] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller. (2022) Sionna. Software package, ver. 1.2.1.
- [19] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, arXiv:1707.06347.
- [20] A. Alkhateeb, G. Charan, T. Osman, A. Hredzak, J. Morais, U. Demirhan, and N. Srinivas, "Deepsense 6g: A large-scale real-world multi-modal sensing and communication dataset," *IEEE Commun. Mag.*, vol. 61, no. 9, pp. 122–128, 2023.
- [21] T. Li, H. Lei, H. Guo, M. Yin, Y. Hu, Q. Zhu, and S. Rangan, "Digital twin-enhanced wireless indoor navigation: Achieving efficient environment sensing with zero-shot reinforcement learning," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 2356–2372, 2025.
- [22] R. Chen, H. Lei, H. Guo, M. Mezzavilla, H. Poddar, T. Yoshimura, and S. Rangan, "Transformer-based rate prediction for multi-band cellular handsets," 2026, arXiv:2509.25722.
- [23] R. Pegurri, F. Linsalata, E. Moro, J. Hoydis, and U. Spagnolini, "Toward digital network twins: Integrating sionna rt in ns-3 for 6g multi-rat networks simulations," in *Proc. IEEE INFOCOM 2025 Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, 2025, pp. 1–6.
- [24] V. Raghavan, M. Kovacevic, R. Ratasuk, A. Ghosh, M. Rybalko, S. Subramanian, S. Hong, A. Tarighat, and Z. Pi, "Spatio-temporal impact of hand and body blockage for millimeter-wave user equipment design at 28 GHz," *IEEE Commun. Mag.*, vol. 56, no. 12, pp. 46–52, 2018.
- [25] V. Raghavan, S. Subramanian, A. Tarighat, Z. Pi, and A. Ghosh, "Hand and body blockage measurements with form-factor user equipment at 28 GHz," *IEEE Trans. Antennas Propag.*, vol. 70, no. 1, pp. 606–619, 2022.
- [26] W. Qi, J. Huang, J. Sun, Y. Tan, C. Wang, and X. Ge, "Measurements and modeling of human blockage effects for multiple millimeter wave bands," in *Proc. IEEE IWCMC*, 2017, pp. 1604–1609.
- [27] U. T. Virk and K. Haneda, "Modeling human blockage at 5G millimeter-wave frequencies," *IEEE Trans. Antennas Propag.*, vol. 68, no. 3, pp. 2256–2266, 2020.
- [28] D. Shakya, M. Ying, T. S. Rappaport, P. Ma, I. Al-Wazani, Y. Wu, Y. Wang, D. Calin, H. Poddar, A. Bazzi, M. Chafii, Y. Xing, and A. Ghosh, "Urban outdoor propagation measurements and channel models at 6.75 GHz FR1 (c) and 16.95 GHz FR3 upper mid-band spectrum for 5G and 6G," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2025, pp. 3291–3296.
- [29] D. Shakya, M. Ying, and T. S. Rappaport, "Angular spread statistics for 6.75 GHz FR1 (c) and 16.95 GHz FR3 mid-band frequencies in an indoor hotspot environment," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2025, pp. 1–6.
- [30] S. Mukherjee, G. Skidmore, T. Chawla, A. Bhardwaj, C. Gentile, and J. Šenić, "Scalable modeling of human blockage at millimeter-wave: A comparative analysis of knife-edge diffraction, the uniform theory of diffraction, and physical optics against 60 GHz channel measurements," *IEEE Access*, vol. 10, pp. 121 720–121 731, 2022.
- [31] F. Fernandes, M. Duits, J. van den Biggelaar, H. Wang, M. Timmers, A. Nur, and G.-J. Janssen, "Hand blockage impact on 5G mmwave beam management performance," *IEEE Access*, vol. 10, pp. 106 032–106 047, 2022.
- [32] M. Ying, D. Shakya, P. Ma, and *et al.*, "Site-specific location calibration and validation of ray-tracing simulator NYURay at upper mid-band frequencies," *npj Wireless Technology*, vol. 2, p. 8, 2026.
- [33] S. Ju, O. Kanhere, Y. Xing, and T. S. Rappaport, "A millimeter-wave channel simulator NYUSIM with spatial consistency and human blockage," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2019, pp. 1–6.
- [34] S. H. A. Shah and S. Rangan, "Multi-cell multi-beam prediction using auto-encoder LSTM for mmwave systems," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 2194–2209, 2022.
- [35] M. Alrabeiah and A. Alkhateeb, "Deep learning for mmwave beam and blockage prediction using sub-6 GHz channels," *IEEE Trans. Commun.*, vol. 68, no. 9, pp. 5504–5518, 2020.
- [36] W. Xu, F. Gao, X. Tao, J. Zhang, and A. Alkhateeb, "Computer vision aided mmwave beam alignment in V2X communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 4, pp. 2699–2714, 2023.
- [37] Y. Yang, F. Gao, X. Tao, G. Liu, and C. Pan, "Environment semantics aided wireless communications: A case study of mmwave beam prediction and blockage prediction," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 7, pp. 2025–2040, 2023.
- [38] M. Yin, T. Li, H. Lei, Y. Hu, S. Rangan, and Q. Zhu, "Zero-shot wireless indoor navigation through physics-informed reinforcement learning," in *Proc. 2024 IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 5111–5118.
- [39] T. Li, H. Lei, M. Yin, and Y. Hu, "Reinforcement learning with physics-informed symbolic program priors for zero-shot wireless indoor navigation," in *Proc. Reinforcement Learning Conf. (RLC)*, 2025.
- [40] H. Lei, H. Guo, T. Svensson, and S. Rangan, "Beyond point estimates: Likelihood-based full-posterior wireless localization," 2025, arXiv:2509.25719.
- [41] C. A. Balanis, *Antenna Theory: Analysis and Design*, 2nd ed. John Wiley & Sons, 1997.
- [42] Y. Hu, M. Yin, S. Rangan, and M. Mezzavilla, "Parametrization and estimation of high-rank line-of-sight MIMO channels with reflected paths," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3808–3822, 2024.
- [43] P. Mogensen, W. Na, I. Z. Kovács, F. Frederiksen, A. Pokhariyal, K. I. Pedersen, T. Kolding, K. Hugl, and M. Kuusela, "LTE capacity compared to the shannon bound," in *Proc. IEEE Veh. Technol. Conf. (VTC Spring)*, 2007, pp. 1234–1238.
- [44] S. Parkvall, E. Dahlman, A. Furuskar, and M. Frenne, "NR: The new 5G radio access technology," *IEEE Communications Standards Magazine*, vol. 1, no. 4, pp. 24–30, 2018.
- [45] W. Ye, X. Hu, S. Sleder, A. Zhang, U. K. Dayalan, A. Hassan, R. A. Fezeu, A. Jajoo, M. Lee, E. Ramadan *et al.*, "Dissecting carrier aggregation in 5G networks: Measurement, QoE implications and prediction," in *Proceedings of the ACM SIGCOMM 2024 Conference*, 2024, pp. 340–357.
- [46] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.